

Exercise Sheet 6

Discrete Mathematics by Hongfei Fu, 2023.10.16

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Note: In the homework, you can rely on Venn diagrams for intuition, but you should write your homework using only formal proofs.

1. List the members of these sets.

c) $\{x | x \text{ is the square of an integer and } x < 100\}$

Answer Area:

$\{1, 4, 9, 16, 25, 36, 49, 64, 81\}$

2. Determine whether these statements are true or false.

c) $\{\emptyset\} \in \{\emptyset\}$ d) $\{\emptyset\} \in \{\{\emptyset\}\}$

Answer Area:

c) False.

d) True.

3. Suppose that A, B , and C are sets such that $A \subseteq B$ and $B \subseteq C$. Show that $A \subseteq C$.

Answer Area:

$\because A \subseteq B$

$\therefore$ For any element $x \in A$, $x \in B$.

$\because B \subseteq C$

$\therefore$ For any element $x \in B$, $x \in C$.

$\therefore$ For any element $x \in A$, $x \in B$, then $x \in C$.

$\therefore A \subseteq C$

4. Find two sets A and B such that $A \in B$ and $A \subseteq B$

Answer Area:

$A := \{\emptyset\}$

$B := \{\emptyset, \{\emptyset\}\}$

5. The defining property of an ordered pair is that two ordered pairs are equal if and only if their first elements are equal and their second elements are equal. Surprisingly, instead of taking the ordered pair as a primitive concept, we can construct ordered pairs using basic notions from set theory. Show that if we define the ordered pair (a, b) to be $\{\{a\}, \{a, b\}\}$, then $(a, b) = (c, d)$ if and only if $a = c$ and $b = d$.

Answer Area:

- Suppose that $(a, b) = (c, d)$, then $\{\{a\}, \{a, b\}\} = \{\{c\}, \{c, d\}\}$.
 $\therefore \forall x \in \{\{a\}, \{a, b\}\}, x \in \{\{c\}, \{c, d\}\}$.
 $\therefore \{a\} \in \{\{c\}, \{c, d\}\}$, then $\{a\} = \{c\}$. Hence $a = c$.
 $\therefore \{a, b\} \in \{\{c\}, \{c, d\}\}$, $a = c$.
 $\therefore \{a, b\} = \{c, d\}$, then $b = d$ and $a = c$.
- Suppose that $a = c, b = d$.
 $\therefore \{\{a\}, \{a, b\}\} = \{\{c\}, \{c, d\}\}$
 $\therefore (a, b) = (c, d)$

In conclusion, $(a, b) = (c, d)$ if and only if $a = c$ and $b = d$.

6. Let A, B , and C be sets. Show that $(A - B) - C = (A - C) - (B - C)$.

Answer Area:

- For any element x , if $x \in (A - B) - C$
 $\therefore x \in A - B$ and $x \notin C$.
 $\therefore x \in A$ and $x \notin B$ and $x \notin C$.
 $\therefore x \in A - C$ and $x \notin B - C$.
 $\therefore x \in (A - C) - (B - C)$.
- For any element x , if $x \in (A - C) - (B - C)$.
 $\therefore x \in A - C$ and $x \notin B - C$.
 $\therefore x \notin B - C$.
 $\therefore x \notin B$ or $x \in B \cap C$.
 $\therefore x \in A - C$.
 $\therefore x \notin C$, so $x \notin B \cap C$
 $\therefore x \in A, x \notin B, x \notin C$
 $\therefore x \in A - B$ and $x \notin C$.
 $\therefore x \in (A - B) - C$

In conclusion, $(A - B) - C = (A - C) - (B - C)$.

7. Show that if A and B are sets, then

b) $(A \oplus B) \oplus B = A$

Note:The **symmetric difference** of A and B , denoted by $A \oplus B$, is the set containing those elements in either A or B , but not in both A and B .

Answer Area:

- For any element x , if $x \in (A \oplus B) \oplus B$
 - Suppose that $x \in A \oplus B$ and $x \notin B$.
 $\therefore x \in A$.
 - Suppose that $x \notin A \oplus B$ and $x \in B$.
 $\therefore x \in A \cap B$.
 $\therefore x \in A$.

Hence, if $x \in (A \oplus B) \oplus B$, then $x \in A$.

- For any element x , if $x \notin (A \oplus B) \oplus B$
 - Suppose that $x \in A \oplus B$ and $x \in B$.
 $\therefore x \in B$ and $x \notin A$.
 - Suppose that $x \notin A \oplus B$ and $x \notin B$.
 $\therefore x \notin A$ and $x \notin B$.

Hence, if $x \notin (A \oplus B) \oplus B$, then $x \notin A$. If $x \in A$, then $x \in (A \oplus B) \oplus B$

In conclusion, $(A \oplus B) \oplus B = A$

8. Determine whether the symmetric difference is associative; that is, if A, B , and C are sets, does it follow that $A \oplus (B \oplus C) = (A \oplus B) \oplus C$?

Answer Area:

Yes, a proof is as follows.

- For any element x , if $x \in A \oplus (B \oplus C)$.
 - Suppose that $x \in A$ and $x \notin B \oplus C$.
 If $x \in B \cap C$, then $x \in A \cap B \cap C$. So $x \notin A \oplus B$ and $x \in C$. Then we have $x \in (A \oplus B) \oplus C$.
 If $x \notin B$ and $x \notin C$, then $x \in A \oplus B$. Then we have $x \in (A \oplus B) \oplus C$.
 - Suppose that $x \notin A$ and $x \in B \oplus C$.
 If $x \in B$ and $x \notin C$, then $x \in A \oplus B$. Then we have $x \in (A \oplus B) \oplus C$.
 If $x \notin B$ and $x \in C$, then $x \notin A \oplus B$. Then we have $x \in (A \oplus B) \oplus C$.

Hence, if $x \in A \oplus (B \oplus C)$, $x \in (A \oplus B) \oplus C$.

- For any element x , if $x \in (A \oplus B) \oplus C$.
 - Suppose that $x \in A \oplus B$ and $x \notin C$.
 If $x \in A$ and $x \notin B$, then $x \notin B \oplus C$. Then we have $x \in A \oplus (B \oplus C)$.
 If $x \notin A$ and $x \in B$, then $x \in B \oplus C$. Then we have $x \in A \oplus (B \oplus C)$.
 - Suppose that $x \notin A \oplus B$ and $x \in C$.
 If $x \in B$ and $x \in A$, then $x \notin B \oplus C$. Then we have $x \in A \oplus (B \oplus C)$.
 If $x \notin A$ and $x \notin B$, then $x \in B \oplus C$. Then we have $x \in A \oplus (B \oplus C)$.

Hence, if $x \in (A \oplus B) \oplus C$, then $x \in A \oplus (B \oplus C)$.

In conclusion, $A \oplus (B \oplus C) = (A \oplus B) \oplus C$.

9. Show that for any sets A and B

$$\mathcal{P}(A) \cap \mathcal{P}(B) = \mathcal{P}(A \cap B)$$

Answer Area:

- For any element x , if $x \in \mathcal{P}(A) \cap \mathcal{P}(B)$:
 $\therefore x \in \mathcal{P}(A)$ and $x \in \mathcal{P}(B)$.
 $\therefore x \subseteq A$ and $x \subseteq B$.
 $\therefore \forall y \in x, y \in A, y \in B$.
 $\therefore \forall y \in x, y \in A \cap B$.
 $\therefore x \subseteq A \cap B$.
 $\therefore x \in \mathcal{P}(A \cap B)$
- For any element x , if $x \in \mathcal{P}(A \cap B)$.
 $\therefore x \subseteq A \cap B$.
 $\therefore \forall y \in x, y \in A \cap B$.
 $\therefore \forall y \in x, y \in A, y \in B$.
 $\therefore x \subseteq A$ and $x \subseteq B$.
 $\therefore x \in \mathcal{P}(A)$ and $x \in \mathcal{P}(B)$.
 $\therefore x \in \mathcal{P}(A) \cap \mathcal{P}(B)$.

In conclusion, $\mathcal{P}(A) \cap \mathcal{P}(B) = \mathcal{P}(A \cap B)$.

10. Show that $A \times \bigcup \mathcal{B} = \bigcup \{A \times X \mid X \in \mathcal{B}\}$

Answer Area:

- For any element (a, b) , if $(a, b) \in A \times \bigcup \mathcal{B}$
 $\therefore a \in A, b \in \bigcup \mathcal{B}$
 $\therefore \exists X_0 \subseteq \mathcal{B}, b \in X_0$.
 $\therefore (a, b) \in A \times X_0$.
 $\therefore (a, b) \in \bigcup \{A \times X \mid X \in \mathcal{B}\}$.
- For any element (a, b) , if $(a, b) \in \bigcup \{A \times X \mid X \in \mathcal{B}\}$
 $\therefore \exists X_0 \in \mathcal{B}, (a, b) \in A \times X_0$.
 $\therefore a \in A, b \in X_0$.
 $\therefore b \in \bigcup \mathcal{B}$.
 $\therefore (a, b) \in A \times \bigcup \mathcal{B}$.